COMPARSION OF VARIOUS PARAMETERS FOR ADDITION USING RIPPLE CARRY ADDER METHODOLGY FOR TWO 12-BIT NUMBER

Parameters	4-bit approximation	5-bit approximation	6-bit approximation	7-bit approximation	8-bit approximation
MEAN ERROR DISTANCE (MED)	3.5	7.5	15.5	31.5	63.5
NORMALISED MEAN ERROR DISTANCE (NMED)	0.00042798	0.000915639	0.001892323	0.003845684	0.007752411
ERROR ORDER (total no. of cases =16777216)	6029312	7602176	8945664	10092544	11071488
ACCURACY	64.0625%	54.6875%	46.679%	39.84375%	34.0087%

PSNR COMPARISON BETWEEN THE IMAGES WITH THREE DIFFERENT FILTERS

PSNR COMPARISON	**BOX KERNEL**			**LAPLACIAN KERNEL**			**SOBEL KERNEL**		
Images comparison →	Original v/s Exact mean	Original v/s approx mean	Exact v/s approx mean	Original v/s Exact mean	Original v/s approx mean	Exact v/s approx mean	Original v/s Exact mean	Original v/s approx mean	Exact v/s approx mean
4-bit approx	29.10 db	28.88	35.41	6.44	6.31	13.42	6.52	6.58	23.40
5 bit approx	29.10 db	27.16	30.34	6.44	6.46	13.33	6.52	6.76	21.43
6 bit approx	29.10 db	23.23	24.25	6.44	6.87	12.89	6.52	7.06	18.11
7 bit approx	29.10 db	17.32	17.53	6.44	6.98	11.38	6.52	7.04	13.67
8 bit approx	29.10 db	13.24	13.33	6.44	6.73	10.00	6.52	7.07	11.64

SYNTHESIS RESULTS OF THE 12-BIT ADDER WITH AND WITHOUT APPROXIMATION USING ZYNQ 7000 FPGA (xc7z020clg484-1)

	Worst pulse width slack	No. of LUTs	No. of SLICEM	No. of Flip-Flops	Static power	Dynamic power
Without approximation	NA	12	5	0	1.036 W	7.925 W
4-bit approximation	NA	12	6	0	1.003W	7.656 W
5-bit approximation	NA	12	6	0	0.961 W	7.569 W
6-bit approximation	NA	12	6	0	0.921 W	7.478 W
7-bit approximation	NA	12	5	0	0.886W	7.394 W
8-bit approximation	NA	12	6	0	0.850 W	7.303 W

SYNTHESIS RESULT OF 12 BIT ADDER USING 90 NM CMOS TECHNOLOGY USING CADENCE

	Cell count	Area (u m^2)	Leakage power (W)	Internal power (W)	Switching power (W)	Total power (W)	Critical path delay
Without approx	13	227	9.8651e-7	5.0894e-6	4.2148e-6	10.291e-6	3040 ps
4- bit approx	22	225.556	1.0308e-6	4.4643e-6	3.9594e-6	9.4544e-6	2272 ps
5-bit approx	24	224.042	1.0462e-6	4.3432e-6	3.9656e-6	9.3552e-6	2024 ps
6-bit approx	27	223.285	1.0658e-6	4.1934e-6	3.9075e-6	9.1677e-6	1774 ps
7-bit approx	30	222.529	1.0855e-6	4.0520e-6	3.8521e-6	8.9896e-6	1524 ps
8-bit approx	33	221.272	1.1051e-6	3.9229e-6	3.8021e-6	8.8302e-6	1275 ps